

## CSA1304-THEORYOF COMPUTATION LAB EXPERIMENTS

7. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$$S \rightarrow A101A, \quad A \rightarrow 0A \mid 1A \mid \varepsilon$$

### AIM

Write a C program to check whether a given string belongs to the Context-Free Grammar (CFG).

### CODE

```
#include <stdio.h>
#include <string.h>

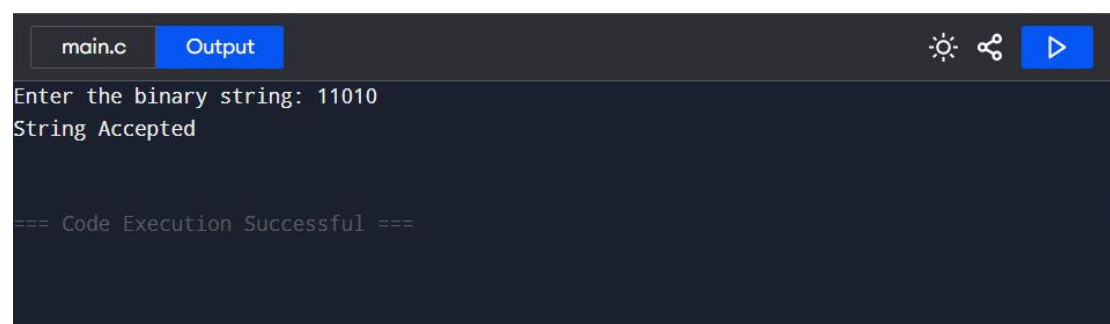
int main()
{
    char str[100];

    printf("Enter the binary string: ");
    scanf("%s", str);

    if(strstr(str, "101") != NULL)
        printf("String Accepted\n");
    else
        printf("String Rejected\n");

    return 0;
}
```

### OUTPUT

A screenshot of a code editor window titled 'main.c' with an 'Output' tab selected. The output shows the program's execution: 'Enter the binary string: 11010' followed by 'String Accepted'. At the bottom, a status message reads '=== Code Execution Successful ==='. The interface includes standard icons for settings, sharing, and running the code.